

Numerical Linear Algebra First-Year Exam, January 2022

Do 4 (four) problems.

1. Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{n \times p}$. Prove or provide a counterexample to the following statements. Justify each nontrivial step.

- (a) The Frobenius norm satisfies $\|AB\|_F \leq \|A\|_F \|B\|_F$.
- (b) $\|A\|_2 \leq \|A\|_F$, where $\|A\|_F$ is the Frobenius norm of A .

2. Define the matrices A and B by

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \\ 2 & 0 & 0 \end{pmatrix}.$$

- (a) Find an economy singular value decomposition of A .
- (b) Find an economy QR decomposition of B .

3. Let $\|\cdot\|$ be a subordinate (induced) matrix norm.

- (a) If E is $n \times n$ with $\|E\| < 1$, then show $I + E$ is nonsingular and

$$\|(I + E)^{-1}\| \leq \frac{1}{1 - \|E\|}.$$

- (b) If A is $n \times n$ invertible and E is $n \times n$ with $\|A^{-1}\|\|E\| < 1$, then show $A + E$ is nonsingular and

$$\|(A + E)^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}\|\|E\|}.$$

4. Let $P \in \mathbb{C}^{m \times m}$ be a projector. Show that $\|P\|_2 = 1$ if and only if P is an orthogonal projector.

5. Suppose A is Hermitian positive definite.

- (a) Prove that each principal submatrix of A is Hermitian positive definite.
- (b) Prove that an element of A with largest magnitude lies on the diagonal.
- (c) Prove that A has a Cholesky decomposition.